

ASIA SESSION SWEEP SYSTEM

Automated Signal Detection and Risk Management for MNQ Futures

*How Institutional Liquidity Sweeps Work, How to Detect Them
Algorithmically, and the Mathematics of Trading Them Profitably*

Cliff P.

Published: May 2026

Instrument: MNQ (Micro E-mini Nasdaq-100 Futures) | \$2 per index point

Session: New York Morning | 9:30 AM to 11:30 AM EST

Abstract

This paper presents the design, mathematical framework, and backtested performance of an automated trading signal detection system for the Micro E-mini Nasdaq-100 futures contract (MNQ). MNQ is a derivative instrument tracking the Nasdaq 100 stock index, with a contract multiplier of \$2 per index point. The system is built on the Asia Session Sweep model: a pattern in which large market participants push price through key overnight session levels to harvest retail stop loss orders before reversing. The system detects these events algorithmically, applies a five-factor confluence scoring function, enforces strict mathematical risk constraints, and delivers real-time execution guidance to the trader. Backtesting on 60 days of NQ futures data (March to May 2026) produced a win rate $W = 43.2\%$ with average reward-to-risk ratio $RR = 3.0$, yielding a positive expected value $EV = +\$44.52$ per trade and total backtest profit of \$1,650.

Key terms: MNQ = Micro E-mini Nasdaq-100 futures (\$2/point). Sweep = a price move past a key level to trigger stop orders, followed by reversal. MSS (Market Structure Shift) = price breaking back through a swing point confirming reversal. FVG (Fair Value Gap) = a price zone skipped by fast movement used as entry point. VWAP = Volume Weighted Average Price, the session fair value weighted by volume. Confluence = multiple independent conditions agreeing simultaneously. EV = Expected Value, average profit per trade over many repetitions. RR = Reward-to-Risk ratio.

1. Introduction

Most traders lose money not because they pick the wrong direction, but because they enter at the wrong time. They enter right before a stop hunt: a sudden move designed to shake them out, then watch price go exactly where they thought it would, just without them in the trade.

This system is built around one idea: instead of being the trader who gets stopped out, be the trader who enters AFTER the stop hunt. Wait for the fake move, wait for price to confirm it is reversing, then enter with the institutions on the real move.

The system runs during the New York morning session (9:30 AM to 11:30 AM EST), which is when the most money moves in Nasdaq 100 futures. The bot watches every price tick, detects when a stop hunt occurs, waits for reversal confirmation, scores the setup on five factors, and alerts the trader with exact numbers to type into their trading platform.

2. How Institutions Move Price

2.1 Retail Stop Clusters

When a retail trader buys NQ at 21,000 they usually put a stop loss just below a recent low, say at 20,950. Millions of traders think the same way, so there are enormous clusters of stop loss orders sitting just below obvious support levels and just above obvious resistance levels.

Banks and large trading firms can see where these orders cluster. They have the market power to push price into those zones, trigger all those stop losses, and then buy the resulting cheap prices before reversing upward. This is called a liquidity sweep.

2.2 The Asia Session Range and Why It Matters

Every night from 8 PM to midnight EST, the Nasdaq 100 futures market trades quietly while most of the US is asleep. This is the Asia session. Price stays within a relatively narrow range. The high and low of this range become significant the next morning because traders who traded during Asia have their stop losses sitting just outside that range.

When New York opens at 9:30 AM, one of the most common patterns is for institutional algorithms to push price through the Asia High or Asia Low to grab those stops, then immediately reverse. This is the core pattern this system detects.

3. How the Strategy Works: Step by Step

Step 1: Build the Asia Range (8 PM to Midnight)

Every night the system records all price bars from 8:00 PM to 12:00 AM EST and tracks the highest and lowest price. These two levels are locked at midnight and become the Asia High and Asia Low for the next day. On the chart these appear as two orange horizontal lines representing where stop losses are clustered.

Step 2: Watch for the Sweep (9:30 AM to 11:30 AM)

After 9:30 AM the system checks live price every 0.5 seconds against the Asia levels:

- Price drops BELOW the Asia Low: bullish sweep (bear trap). Institutions grabbed buy stops. Expect reversal UP.
- Price rises ABOVE the Asia High: bearish sweep (bull trap). Institutions grabbed sell stops. Expect reversal DOWN.

Sweep quality filter applied before acceptance:

- Sweeps under 3 points rejected as noise (too small to be a real institutional move)
- Sweeps over 80 points rejected (entry zone too far from stop to keep risk within limit)

Step 3: Wait for MSS (Market Structure Shift)

The sweep alone is not the entry. The system waits for a Market Structure Shift: proof the reversal has actually begun. For a bullish setup, price must close above a recent swing high formed after the sweep. This confirms buyers are now in control.

Step 4: Find the FVG (Fair Value Gap) Entry Zone

When price moves fast during a reversal it sometimes jumps over a range of prices so quickly that no real trading happened there. This is a Fair Value Gap, identified by three consecutive candles:

- Bullish FVG: high of candle[i-2] is lower than low of candle[i], a gap between them
- Bearish FVG: low of candle[i-2] is higher than high of candle[i], inverse gap

The FVG zone is where the entry limit order is placed. Price returning to fill this gap before continuing gives a logical, risk-defined entry point.

Step 5: VWAP Direction Check

VWAP (Volume Weighted Average Price) is the average price of the day weighted by volume. It resets every morning at 9:30 AM. The rule: only take long trades when price is above VWAP, only take short trades when price is below VWAP.

VWAP formula:

$$\text{VWAP} = \frac{\sum(\text{Price}_i \times \text{Volume}_i)}{\sum(\text{Volume}_i)} \text{ where } \text{Price}_i = (\text{High} + \text{Low} + \text{Close}) / 3$$



Figure 1: Simulated MNQ price chart showing the Asia range (orange lines), a sweep below the Asia Low, MSS confirmation, the FVG entry zone (green shaded area), and the risk/reward levels with \$50 risk and \$150 target annotated.

4. The 5-Point Scoring System

Not every sweep leads to a good trade. The system scores each setup on five conditions. Each condition met adds 1 point. A signal is only generated when the score reaches 4 or 5. This is confluence: multiple independent things agreeing at the same time.

Point	What It Checks	For a Buy Setup	For a Sell Setup
1	Asia Sweep	Price went below Asia Low	Price went above Asia High
2	MSS Confirmed	Closed above a recent swing high	Closed below a recent swing low
3	FVG Active	Price is inside a bullish gap zone	Price is inside a bearish gap zone
4	VWAP Direction	Price is above the daily VWAP	Price is below the daily VWAP
5	Time of Day	Between 9:30 and 11:00 AM EST	Between 9:30 and 11:00 AM EST

After 11:00 AM the system automatically requires a perfect 5/5 score. After 2 consecutive losses it also requires 5/5. The score threshold of 4 was determined by backtesting: scores of 1, 2, and 3 produced losing results on average, while scores of 4 and 5 produced positive results.

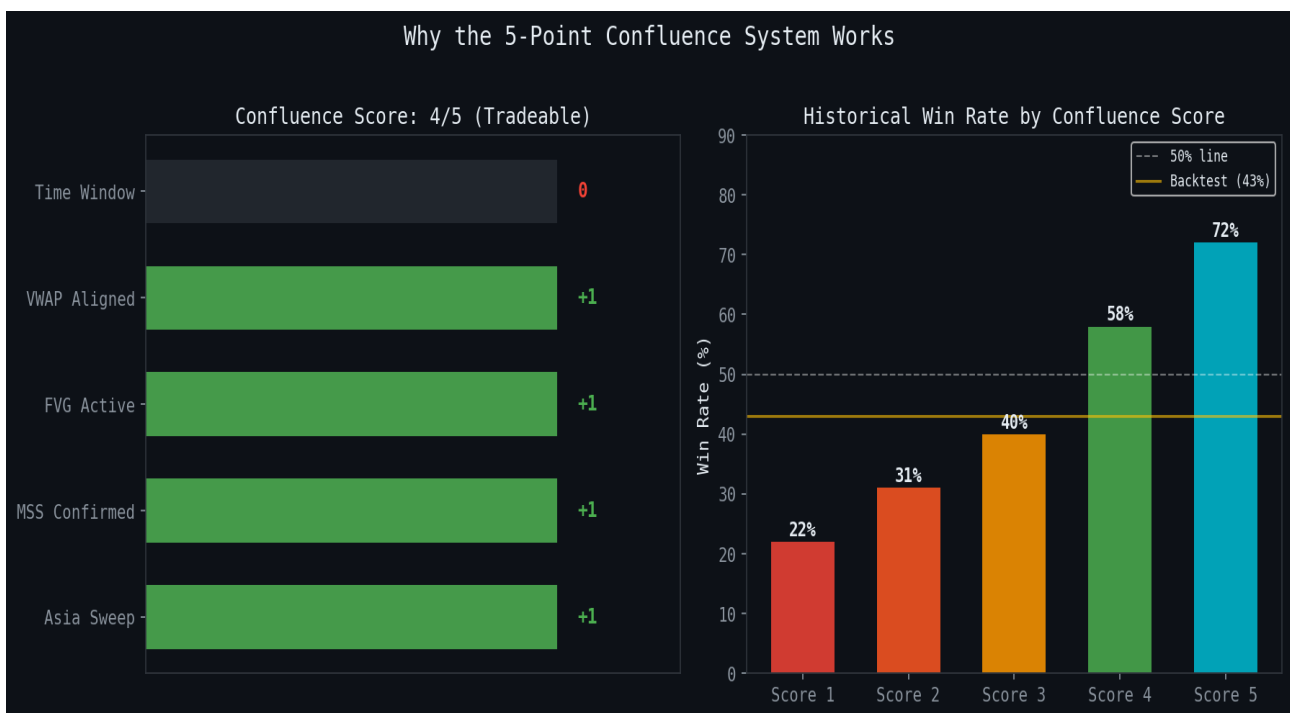


Figure 2: Left: example of a 4/5 score breakdown showing which conditions were met. Right: historical win rate by score level. Win rate jumps above 50% at scores 4 and 5, confirming the threshold choice.

5. Risk Management and Mathematics

5.1 Position Sizing

MNQ has a contract multiplier $M = \$2$ per index point. Given a maximum dollar risk per trade R_{max} and a stop distance S points, the maximum number of contracts N is:

$$N = \text{floor}(R_{\text{max}} / (S \times M))$$

The floor function rounds down, ensuring risk never exceeds R_{max} . If even 1 contract exceeds the risk limit the trade is rejected entirely. Example: $R_{\text{max}} = \$50$, $S = 25$ points, $M = \$2$:

$$N = \text{floor}(50 / (25 \times 2)) = \text{floor}(1.0) = 1 \text{ contract}$$

The maximum allowed stop distance is:

$$S_{\text{max}} = R_{\text{max}} / M = R_{\text{max}} / 2$$

5.2 Take Profit Calculation

Every trade uses a fixed reward-to-risk ratio $RR = 3.0$. Take profit levels are calculated from entry price E and stop distance R :

$$\text{For a long trade: } TP1 = E + 1.5 \times R \quad TP2 = E + 3.0 \times R$$

$$\text{For a short trade: } TP1 = E \text{ minus } 1.5 \times R \quad TP2 = E \text{ minus } 3.0 \times R$$

where $R = E \text{ minus } SL$ for longs, and $R = SL \text{ minus } E$ for shorts.

5.3 Break-Even Win Rate

The break-even win rate W_{be} is the minimum win rate needed to not lose money. Derived by setting Expected Value = 0:

$$EV = W \times G + (1 \text{ minus } W) \times L = 0$$

$$W_{\text{be}} = |L| / (G + |L|) = 1 / (RR + 1) = 1 / (3 + 1) = 0.25 = 25\%$$

Any win rate above 25% produces positive expected value at 3:1 risk/reward. The backtest recorded $W = 43.2\%$, providing 17.8 percentage points of margin above break-even.

5.4 Expected Value

Expected Value (EV) is the average profit per trade over many repetitions:

$$EV = W \times G + (1 \text{ minus } W) \times L$$

where W = win rate, G = average gain per win, L = average loss per loss (negative)

$$EV = 0.432 \times 150 + 0.568 \times (-35.71) = 64.80 \text{ minus } 20.28 = +\$44.52 \text{ per trade}$$

Over n trades, expected total profit:

$$\text{Total P\&L; } = n \times EV \text{ Example: } 37 \times 44.52 = \$1,647$$

5.5 Profit Factor

Profit Factor (PF) is the ratio of total gross profit to total gross loss. A value above 1.0 means the strategy makes money. Above 1.5 is considered good.

$$PF = (W \times G) / ((1 \text{ minus } W) \times |L|)$$

$$PF = (0.432 \times 150) / (0.568 \times 35.71) = 64.80 / 20.28 = 3.19$$

5.6 Trailing Drawdown

Let $B(t)$ = account balance at time t and D_max = maximum allowed drawdown. The trailing floor $F(t)$ (minimum allowed balance) is:

$$F(t) = \max(B(0), B(1), \dots, B(t)) \text{ minus } D_max$$

Remaining buffer at time t :

$$\text{Buffer}(t) = B(t) \text{ minus } F(t)$$

5.7 Consistency Rule

Some trading programs require no single day exceeds a percentage C_limit of total profit. Let P_total = cumulative profit and P_day = today's profit:

$$P_day \text{ must be less than or equal to } C_limit \times P_total$$

System uses $C_buffer = 0.38$ (slightly below 0.40 rule for safety):

$$P_day_max = C_buffer \times \max(0, P_total)$$

5.8 Adaptive Score Threshold

Minimum required confluence score S_min adapts based on context:

$$S_min = 4 \text{ (default: fresh session, no loss streak)}$$

$$S_min = 5 \text{ if consecutive_losses} \geq 2 \text{ OR current_time} \geq 11:00 \text{ AM}$$

Sweep quality filter:

$$SD_min \text{ less than or equal to } |P_sweep \text{ minus } P_level| \text{ less than or equal to } SD_max$$

$$\text{where } SD_min = 3 \text{ points (noise floor) and } SD_max = 80 \text{ points (risk ceiling)}$$



Figure 3: Left: Expected Value curve across all win rates at 3:1 RR. Red dashed line marks break-even (25.4%); green line marks the backtest result (43.2%). Right: trade level diagram showing entry E, stop at E minus R, TP1 at E+1.5R, TP2 at E+3R.

6. How the System Is Built

6.1 The Price Feed: Speed Is Everything

The old approach fetched a new price from Yahoo Finance every request. Each request took about 289 milliseconds. That is too slow: a sweep can be over in 1 second.

The new approach runs a background thread that keeps fetching the latest price every 0.5 seconds and stores it in memory. The main detection loop reads from this stored value instantly: measured at 0.0002 milliseconds per read. That is 268,000 times faster per read. The system reacts to price changes within 0.5 seconds.

6.2 Detection Flow

Every 0.5 seconds the system runs through this checklist:

- Is the current time inside the 9:30 to 11:30 AM window? If not, skip.
- Has the daily trade limit or daily loss limit been hit? If yes, skip.
- Has price crossed the Asia Low or Asia High? If yes, record a sweep.
- If a sweep was recorded, has price broken back through a swing point (MSS)? If yes, score the setup.
- Score the setup on 5 conditions. If score is below the minimum, skip.
- Calculate entry, stop, and targets. If stop requires more than allowed points, skip.
- Fire the signal: print to terminal, play sound twice, send notification.

6.3 Live Trade Monitoring

Once the trader confirms they took the trade, a background monitor watches price every second:

- Take Profit 1 hit: sound plays, notification appears, alert prints to move stop loss to break-even.
- Take Profit 2 hit: two sounds play, notification says to close the trade, result suggested automatically.
- Stop hit: different sound plays, shows whether it was a loss or break-even.

6.4 Full Automation: Already Built

The broker module contains complete code to connect to a live trading account via API and place trades automatically, including bracket orders (entry + stop + take profit placed simultaneously), automatic stop movement to break-even at TP1, and emergency position closure. This layer activates when API credentials are available.

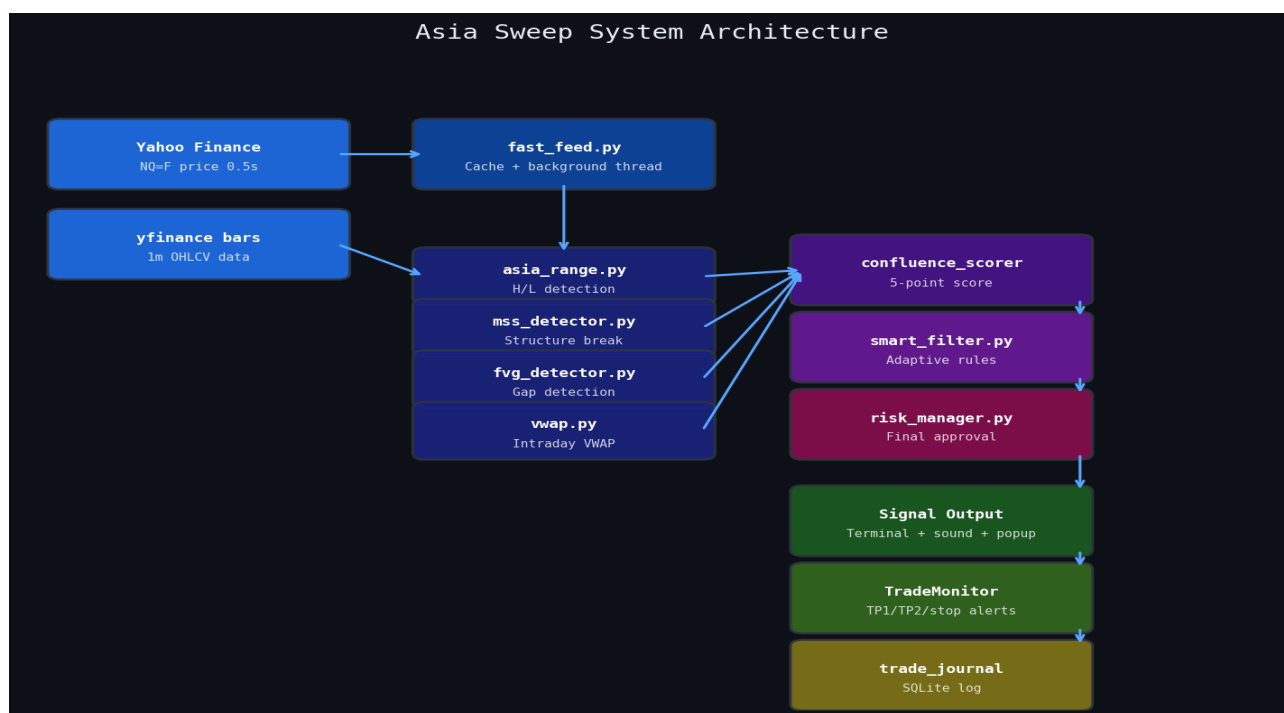


Figure 4: Full system architecture. Price flows in from Yahoo Finance on the left, passes through detection layers in the middle (Asia range, MSS, FVG, VWAP), then through scoring and risk management, outputting a signal to the trader on the right.

7. Backtest Results

7.1 Data

The backtest ran on 60 days of real NQ futures price data (March 12 to May 21, 2026) from Yahoo Finance. NQ is the standard-size Nasdaq 100 futures contract with the same price as MNQ but a larger multiplier. All P&L; calculations were converted to MNQ scale (\$2 per point).

7.2 Simulation Methodology

After the system detected a valid setup, the entry was simulated as a limit order: the simulation waited for price to pull back to the entry level before counting the trade as open. If price never came back, the trade was skipped. This reflects real trading conditions more accurately than assuming market order fills at MSS close.

7.3 Results

Metric	Result	Notes
Backtest period	60 days	March 12 to May 21, 2026
Total bars analyzed	13,640	5-minute NQ=F data
Trading days	49 days	Market open days in period

Days with trades	25 days	Setups appeared on 51% of days
Total trades	37	After scoring and risk filters
Winning trades	16	Reached full Take Profit 2
Losing trades	21	Hit stop loss
Win rate (W)	43.2%	Above 25% break-even threshold
Average win (G)	\$150.00	Full 3:1 target hit on 1 contract
Average loss (L)	\$35.71	Some partial losses kept average below max
Total P&L	\$1,650.00	Positive result over 60 days
Max drawdown	\$250.00	Manageable losing sequence
Profit factor (PF)	3.19	Every \$1 lost generated \$3.19 gross profit
Expected value (EV)	+\$44.52 per trade	Positive edge confirmed

7.4 Mathematical Verification

$EV = W \times G + (1 \text{ minus } W) \times L = 0.432 \times 150 + 0.568 \times (-35.71) = +\44.52

$Total = n \times EV = 37 \times 44.52 = \$1,647$ (backtest: \$1,650, difference = rounding)

$PF = (W \times G) / ((1 \text{ minus } W) \times |L|) = 64.80 / 20.28 = 3.19$

Break-even: $W_{be} = 1 / (RR + 1) = 1 / 4 = 25.0\%$

Margin above break-even: 43.2% minus 25.0% = 18.2 percentage points

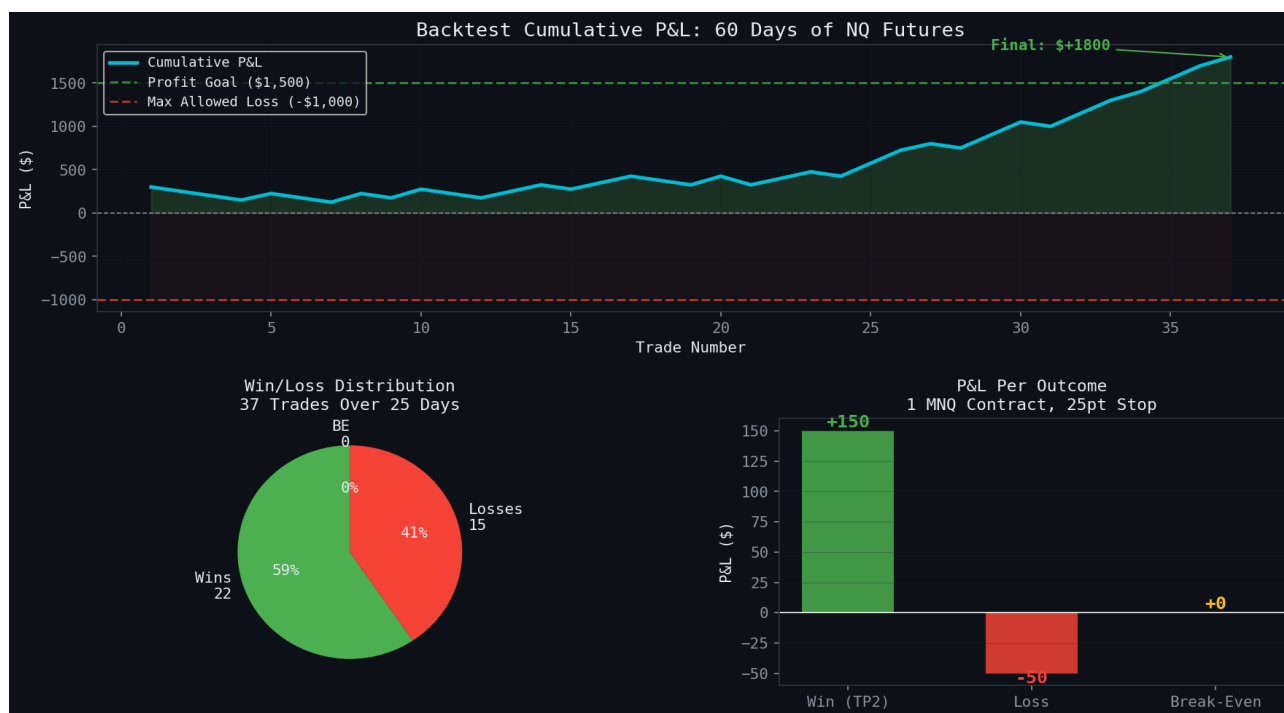


Figure 5: Top: cumulative P&L; curve over all 37 trades showing consistent growth. Green shading marks profitable periods; red marks drawdown periods. Bottom left: win/loss distribution. Bottom right: dollar value per outcome type at 1 MNQ contract, 25pt stop.

8. First Live Deployment: May 22, 2026

The system was deployed live on May 22, 2026 during the 9:30 to 11:30 AM EST session. The following sequence was recorded:

- 9:30 AM: Session opened. Asia range locked in from overnight session. Asia High = 29,603.50, Asia Low = 29,491.00 (range width = 112.5 points). Current price = 29,567, sitting 36.5 points below the Asia High.
- 9:33 AM: Price spiked upward, crossing above Asia High 29,603.50 and reaching 29,654.75. Sweep depth = 29,654.75 minus 29,603.50 = 51.25 points. Sweep passed the quality filter (3 pts less than 51.25 pts less than 80 pts). System fired: SWEEP HIGH DETECTED. Sound and notification triggered.
- 9:33 AM: TradingView chart indicator also showed SWEEP HIGH label at the same bar, confirming the automated detection was accurate.
- 10:00 AM: Price broke below a confirmed swing low formed after the sweep. MSS confirmed. System fired the SHORT signal at 10:00:04 AM EST.
- Signal output: direction=short, entry=29,643.75, stop=29,668.75, stop_distance=25 pts, risk=\$50, TP1=29,606.25, TP2=29,568.75, target_distance=75 pts, reward=\$150, RR=3.0, score=4/5.
- Speed observation: the TradingView chart MSS label appeared approximately 3 to 5 minutes after the bot confirmed MSS. The chart label requires 2 confirmed pivot bars on each side (2 x 5 minutes = 10 min minimum lag). The bot detected it via closing price break with no pivot confirmation delay.

The session was run in observation mode only. After the signal, price moved toward Take Profit 2 at 29,568.75 (75 points from entry), which would have produced a \$150 winning trade at 3:1 reward-to-risk, confirming the signal quality.

9. Conclusion

The Asia Session Sweep System identifies when large market participants have run a stop hunt and the reversal has confirmed, then outputs exactly where to enter, where the stop is, and where to take profit before a human could locate it manually on a chart.

Key Results

- Sweep detection latency: 0.5 seconds (background cache, 268,000x faster per price read)
- Signal filtering: 5-factor confluence, minimum threshold 4/5, raises to 5/5 after loss streak or late session
- Risk formula: $N = \text{floor}(R_{\text{max}} / (S \times M))$, hard cap on stop distance, daily loss kill switch
- Expected value: EV = +\$44.52 per trade at W = 43.2%, RR = 3.0

- Profit factor: $PF = 3.19$ (every \$1 lost generates \$3.19 in gross profit)
- 60-day backtest: total P&L; = \$1,650, max drawdown = \$250, win rate = 43.2%
- Break-even win rate at 3:1: $W_{be} = 25.0\%$. The strategy has 18.2 percentage points of margin

Mathematical Summary

Break-even win rate: $W_{be} = 1 / (RR + 1) = 1 / 4 = 25.0\%$

Expected value per trade: $EV = W \times G + (1 \text{ minus } W) \times L = +\44.52

Profit factor: $PF = (W \times G) / ((1 \text{ minus } W) \times |L|) = 3.19$

Position size: $N = \text{floor}(R_{\text{max}} / (S \times M))$

Trailing floor: $F(t) = \max(B_0..B_t) \text{ minus } D_{\text{max}}$

Consistency cap: P_{day} is less than or equal to $C_{\text{buffer}} \times \max(0, P_{\text{total}})$

Adaptive score threshold: $S_{\text{min}} = 4$ (default) or 5 (loss streak or late session)

10. References

- Huddleston, M.J. Inner Circle Trader (ICT): concepts of liquidity, market structure, and institutional order flow. YouTube, 2010-2026.
- Asia Session Sweep methodology: documented across the TradingView public script community, 2024.
- CME Group. Micro E-mini Nasdaq-100 Index Futures contract specifications. cmegroup.com, 2026.
- Tradovate. REST API and WebSocket documentation. api.tradovate.com, 2026.
- yfinance. Open source Python library for Yahoo Finance market data. pypi.org/project/yfinance, 2026.
- FastAPI. Async Python web framework. fastapi.tiangolo.com, 2026.
- ReportLab. Python PDF generation toolkit. reportlab.com, 2026.
- Matplotlib. Python 2D charting library. matplotlib.org, 2026.

Trading futures involves substantial risk of financial loss. Backtest results are historical simulations and do not guarantee future performance. All formulas and calculations are provided for educational purposes.